Learning Eigenstructures of Unstructured Data Manifolds

We sincerely thank the reviewers for their time and valuable feedback. We address main concerns below.

Fixed Point Cloud Size. Reviewer 39zS states "the method requires all point clouds to have the same number of points." Incorrect: our Transformer handles variable-length sequences. Fixed training size is for batching efficiency—the model generalizes to different point counts at inference. Concretely, the same overfitted Pegaso model (trained on 4k points) achieves mean cosine similarity 0.606 at 3k points and 0.661 at 12k points (Fig. 21, 12th basis function). This holds across all shapes. Similarly, we train on 5000 image samples but evaluate on 1500 in Figs. 7–8.

Figure 21. 3k vs. 12k points.

Why Main Focus on Laplacians? While our theory is operator-agnostic, we focus on smooth probes (yielding Laplacian-like bases) as this alone opens substantial research directions. In high dimensions, discretizing a metric-aware Laplacian is unclear—our approach sidesteps this. In 3D, learning task-specific spectral bases (e.g., for segmentation, with reconstruction loss as regularization) is a compelling direction.

Additional Probe Experiments. We conducted experiments showing different probe distributions recover different spectral bases.

Figure 22. Probe (left), basis (right).

(A) Piecewise-constant probes: Using piecewise-constant functions on K clusters as probes—mimicking semantic priors (e.g., part labels)—we observe two regimes: when predicting fewer eigenfunctions than clusters ($k < K$), the network converges to a Walsh-Hadamard-like basis (Fig. 22); when $k = K$, it converges to indicator functions. *(B) Polynomial probes:* Using random polynomials of degree ≤ 2 (a 10D space spanned by $\{1, x, y, z, x^2, y^2, z^2, xy, xz, yz\}$), the network converges to a basis spanning the same subspace (99.93% variance explained vs. 0.24% for a random baseline), despite polynomials being smooth like Laplacian eigenfunctions. This confirms *uniformly random smooth probes* are crucial—Gaussian-smoothed noise samples uniformly from the Laplacian's low-energy eigenspace. *(C) Schrödinger-like probes:* Modulating probes by a potential proportional to Gaussian curvature before smoothing, the recovered basis resembles eigenfunctions of the Hamiltonian $H = \Delta + \beta V$ (Fig. 23)—the Laplacian's natural generalization with a potential. These confirm: *different probes yield different spectral bases.*

Downstream Applications. We focused and elaborated on motivation, theory, and validating fundamentals—spectral basis fidelity in 3D and high-dimensional applicability. Thorough downstream evaluation requires dedicated work. Promising directions: *(A)* DiffusionNet [Sharp et al., ACM TOG 2022], replacing its fixed Laplacian basis with ours; *(B)* Latent Functional Maps [Fumero et al., NeurIPS 2024], replacing graph Laplacian eigenvectors; *(C)* Feature compression: FeatUp [Fu et al., ICLR 2024] uses PCA—our bases could be a data-adaptive alternative; *(D)* Scalable dimensionality reduction (faster than UMAP, any k).

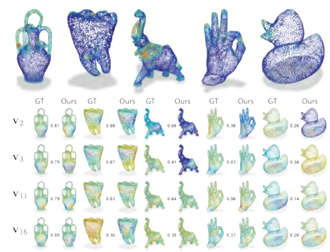

Figure 23. potential $V(x)$ (top), learned vs. GT (bottom).

Probe Functions Design & Hyperparameters. Reviewer Ff1E notes "the burden is passed on to probe function design." This is a *favorable* trade-off: smoothing noise via k-NN diffusion is simpler than discretizing operators, especially in high dimensions. Precise tuning is not necessarily required: our 3D experiments use *identical* parameters across all shapes (supp. lines 1388–1392). Regarding sensitivity (3vpu, 39zS): this primarily affects *eigenvalue recovery, not the spectral basis*. Fig. 24 shows different hyperparameters all yield bases closely resembling cotangent Laplacian eigenfunctions. A "wildcard" setting—randomly sampling hyperparameters per probe—achieves competitive cosine similarity, alleviating the need for precise tuning. Table 2's per-shape tuning shows recovering *exact* cotangent Laplacian eigenvalues requires hyperparameter search. Without tuning, the basis still exhibits correct spectral decay (Fig. 20) and may benefit downstream tasks where exact recovery is unnecessary (untested). Probe hyperparameters could also be made learnable w.r.t. a downstream task.

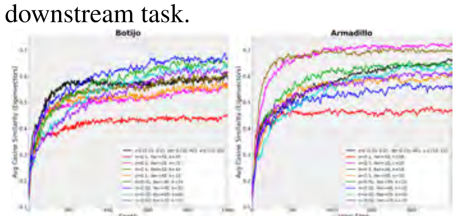

Figure 24. Hyperparameter sensitivity.

Computational Cost. *3D overfitting:* ∼8 min on RTX 3090 (4k points); overfitted models generalize to other samplings. *Inference (50 eigenvectors):* 0.44s/1.14s/4.20s at 3k/12k/24k points—comparable to robust Laplacian [Sharp & Crane, CGF 2020] (0.20s/1.24s/2.62s). We applied no optimization (torch.compile, flash attention, custom CUDA kernels); inference time could be substantially improved, especially with batching. *Generalization:* 24H on 8 A100s (one-time); inference is a single forward pass. *High-dim:* small manifolds (∼1500 samples) run on single RTX 3090.